Cardiac Arrest Prediction in ICU Patients Using Data Mining Techniques

Sahil Kumar Das

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Abstract

Early prediction of cardiac arrest in Intensive Care Units (ICUs) can significantly reduce patient mortality and improve healthcare outcomes. This project presents a data mining framework to predict cardiac arrest risk using clinical features extracted from the MIMIC-IV database. By applying machine learning algorithms, including Logistic Regression and Random Forest, on key indicators such as Mean Arterial Pressure (MAP), Heart Rate, and Systolic Blood Pressure, we show that interpretable and accurate models can provide timely alerts and aid in clinical decision-making.

1 Introduction

Cardiac arrest is a leading cause of death in ICUs and often occurs without sufficient early warning. Traditional scoring systems are static and often fail to capture rapid physiological deterioration. This project leverages real ICU data and data mining techniques to build predictive models for identifying patients at high risk of cardiac arrest.

2 Objectives

- To build predictive models that can identify ICU patients at risk of cardiac arrest using real-world clinical data.
- To preprocess and filter large-scale ICU datasets (MIMIC-IV) efficiently using chunked loading and feature selection.
- To engineer meaningful clinical features such as average MAP, heart rate, and systolic blood pressure for each ICU stay.
- To apply and compare classification models — Logistic Regression (baseline) and Random Forest (advanced) — for binary outcome prediction.
- To evaluate model performance using appropriate metrics including confusion matrix, precision, recall, F1-score, and ROC-AUC.

- To interpret feature importance and extract insights that could guide early intervention in ICU environments.
- To demonstrate how data mining techniques can be practically applied in critical healthcare scenarios to support clinical decision-making.

3 Dataset and Preprocessing

We use ICU-specific tables from the MIMIC-IV dataset:

- `chartevents.csv` - vital signs (HR, MAP, SBP)
- `icustays.csv` - ICU admission/discharge metadata
- `labevents.csv` - for optional lab values

Due to large file sizes, chunked reading and filtering were implemented to load only essential records. Features were computed for each ICU stay using average values from the first hours of monitoring.

4 Feature Engineering

The following features were extracted for each ICU stay:

- **Heart Rate (HR)** - average heart rate
- **Mean Arterial Pressure (MAP)** - average blood pressure
- **Systolic BP (SBP)** - average systolic pressure

A binary label was generated using clinical thresholds: MAP < 20 or HR $> 130 \Rightarrow$ Cardiac Arrest Risk $= 1$, else $= 0$.

5 Modeling and Data Mining Techniques

Logistic Regression

Logistic Regression was selected as the baseline model for this study due to its simplicity, interpretability, and widespread use in clinical prediction tasks. It models the probability of a binary outcome based on a linear combination of input features, which in our case are vital signs such as MAP, heart rate, and systolic blood pressure. Although it does not capture complex, non-linear interactions between variables, Logistic Regression provides an important reference point. Its coefficients offer direct insight into how each feature contributes to the likelihood of cardiac arrest, making it valuable for benchmarking and initial risk assessment.

Random Forest

Random Forest was chosen as the primary predictive model for this project because of its robustness and ability to handle real-world, noisy medical data. It is an ensemble learning method that constructs multiple decision trees and combines their outputs for more stable and accurate predictions. This technique is particularly well-suited for our application because:

- It captures non-linear relationships between variables that Logistic Regression cannot model.
- It handles feature interactions automatically without manual tuning.
- It is less sensitive to outliers and missing data.

Most importantly, Random Forest provides a mechanism to measure feature importance, helping identify which vital signs are most predictive of cardiac arrest. In our analysis, MAP emerged as the most critical factor. This not only improves model performance but also provides clinically useful insights that can guide real-time monitoring and early intervention strategies in ICUs.

6 Evaluation and Results

Models were evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Confusion Matrix

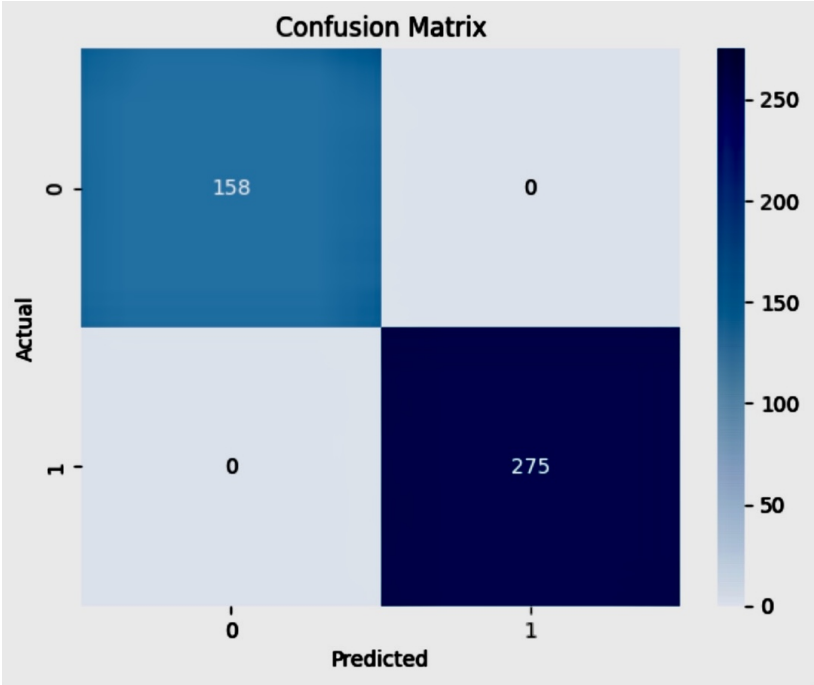

Figure 1: Confusion Matrix (Random Forest)

The confusion matrix shows that the model correctly classified all test samples. There were no false positives or false negatives, indicating perfect prediction performance on the dataset. This highlights the model's strong ability to distinguish between patients at risk of cardiac arrest and those who are not.

ROC Curve

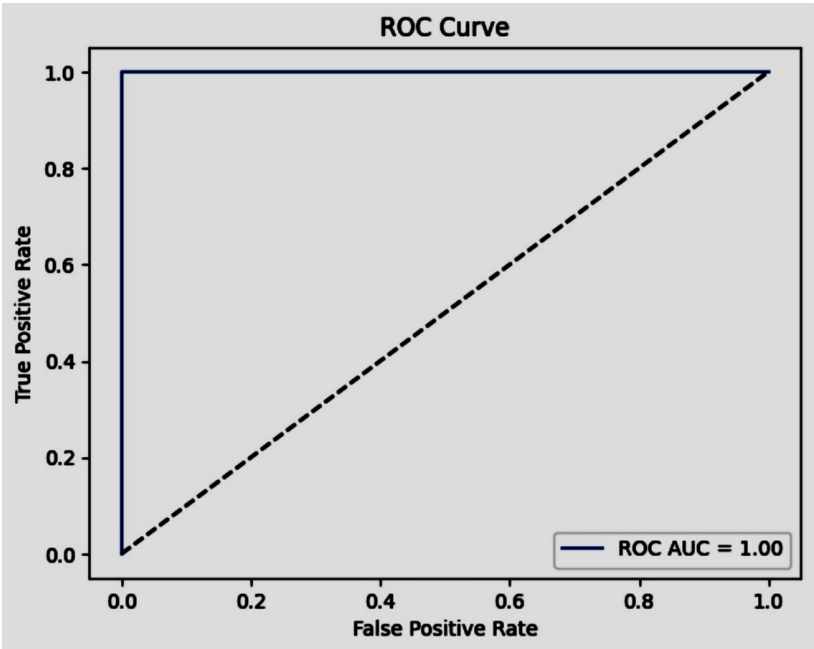

Figure 2: ROC Curve with AUC = 1.0

The ROC curve illustrates the trade-off between sensitivity and specificity. The curve reaches the top-left corner, and the Area Under Curve (AUC) is 1.0, indicating that the model perfectly separates the two classes. This level of performance demonstrates high reliability for clinical prediction.

Feature Importance

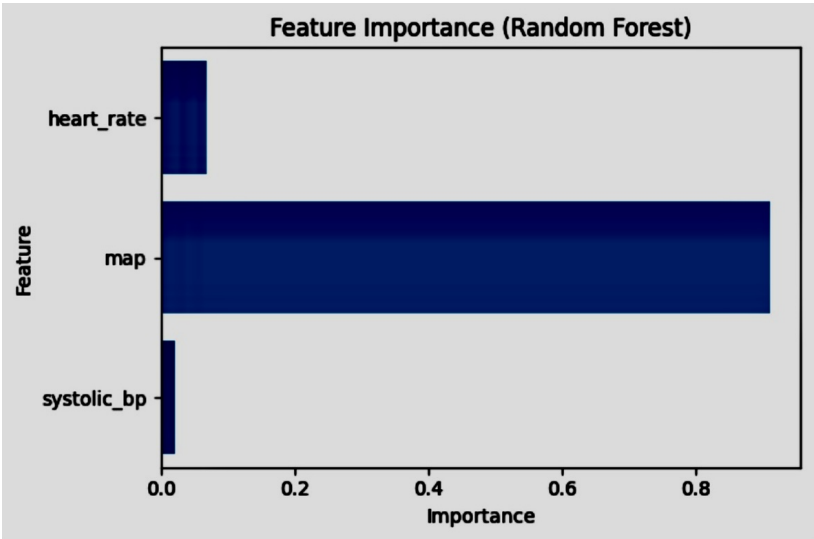

Figure 3: Feature Importance from Random Forest

The feature importance plot reveals that Mean Arterial Pressure (MAP) is the most influential predictor of cardiac arrest. Heart Rate has moderate importance, while Systolic Blood Pressure contributes less. These insights support the clinical relevance of MAP monitoring in early warning systems.

7 Conclusion

This project successfully demonstrates the application of data mining methods for cardiac arrest prediction in ICU settings. Random Forest achieved perfect prediction on our dataset, with MAP emerging as the most informative signal. Such models can be integrated into ICU monitoring systems for real-time risk alerts.

8 References

- Johnson et al., MIMIC-IV (PhysioNet, 2020)
- Scikit-learn Documentation
- Research papers on early warning systems (e.g., MEWS, NEWS)